# THE NEW LOOP

## A FUTURE HISTORY OF THE AI TRANSITION

Economic discussions of the productivity transition driven by artificial intelligence and advanced automation have largely proceeded through uncoupled, qualitative, topical debates. Proposals such as Universal Basic Income are discussed in isolation and with limited structural context. Yet the transition is bound to involve many simultaneous, interacting pressures: a declining share of income flowing through labor, changing patterns of household demand, rising concentration of capital income, fiscal strain, and risks of debt-deflation and political instability. The interactions of such forces will help determine our future, but cannot be fully understood through single-policy arguments.

The future-history narrative in this paper describes a framework of mutually supportive policy mechanisms that interact over time within a single economic system. It aims to make the *structure* of the transition explicit and understandable, showing how a coordinated policy package can address several dimensions of the problem simultaneously and providing a basis for more disciplined public discussion. It does not claim to be "the answer"; it is purposeful storytelling.

The scenario brings together a Public Trust Fund, household income support, a consumption tax with low-income rebates, distribution-sensitive income taxation, automation-responsive corporate tax levels, and mortgage adaptation mechanisms. Taken together, these policies form a coordinated architecture rather than a collection of isolated ideas. A companion paper, *New Loop Policy Architecture and Simulator*, develops the policy design in greater detail and presents a parameterized Stock-Flow Consistent macroeconomic simulation, with illustrative output graphs, along with a reading list of relevant publications.

One secret of scenario-building is that, like *The Lord of the Rings*, a good story must believe in itself. Good modeling, in contrast, is about exploration. A public Streamlit implementation of the *New Loop* model allows readers to inspect assumptions, adjust parameters, and test outcomes for themselves. I hope this work encourages better debate by making the *structure* of the coming economic transition explicit, visible, and open to quantitative testing.

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## INTRODUCTION

Public discussion of advanced artificial intelligence has concentrated on technical risk, safety, and alignment. Those concerns are real. But profound social effects will emerge in the structure of the economy itself, as automation changes how income is earned, how demand is sustained, how capital is rewarded, and how social stability is maintained. Despite its urgency, this transition is only beginning to receive the attention it deserves.

Historically, the revenues earned from economic activity are split among labor, capital, and government. In plain terms: consumers buy goods and services; businesses use that revenue to pay workers, pay taxes, and generate profits; workers then spend and save; owners invest and consume out of profits; and the cycle continues.

This is the "Old Loop" of capitalism: a circular feedback system that maintains balance because the capacity to produce is meaningless without the ability to buy. Wages are not the only source of purchasing power - people can also consume out of dividends, transfers, credit, and asset income - but the system must reliably generate broad, spendable income at the scale required to clear the economy's output.

The consequences can be severe and persistent when that loop fails. Collapsing demand propagates into collapsing production: falling sales lead to layoffs and cutbacks; layoffs reduce incomes; reduced incomes further reduce sales; investment stops; and productive capacity can be impaired for years. The Great Depression was a feedback failure so profound that even extraordinary policy efforts struggled to restore normal economic activity. My grandfather, a railroad fireman who shoveled coal into steam engines, was out of work for a decade. He traded a solid gold railroad watch for a milk cow to plow the garden and feed the family. Whatever one emphasizes as proximate causes, the durable lesson is that restart can be difficult once the link between output and purchasing power is damaged, and the consequences can spill into conflict and politics for generations.

Advanced artificial intelligence is an unusually powerful form of automation. It will automate information and intellectual work first. As that proceeds, it will somewhat later also accelerate automation in physical systems, mining, manufacturing, logistics, agriculture, construction, and beyond. Automation raises productivity and lowers operating costs. But it also changes the income architecture of the economy: as software and machines substitute for labor, a smaller fraction of value added shows up as wages and a larger fraction shows up as profits and returns to capital. If ownership of that capital is concentrated, then purchasing power becomes concentrated too, even while productive capacity expands.

Corporations are the engines of modern wealth creation. Only large-scale enterprises can reliably perform research and development, design, logistics, manufacturing, marketing, distribution, and global support for complex products and services. So the issue is not to stop progress, nor to replace productive enterprise with circular services in which we merely transact among ourselves.

The issue is how the feedback loop that connects production to purchasing power can remain stable as labor income shrinks, so that wealth creation can continue.

That's the core conundrum. If wages stop being a dominant channel by which most households receive income, where does broad purchasing power come from? If we want a prosperous society that continues to invent, build, and improve while much of the operational work is automated, then we will have to redesign parts of the income-and-ownership system, not just "patch" it.

This is not fundamentally a left-right political argument. It's a systems-engineering problem in the design of a modern economy. The timing of change is uncertain; the direction is not. And the failure modes are reasonably clear.

One failure mode is a concentration trap: the rents from automation accrue to narrow ownership, purchasing power becomes too concentrated, demand weakens relative to capacity, and the system's social legitimacy erodes. Another is a treadmill trap: we attempt to offset structurally declining labor income with ad hoc transfers and stimulus without changing the underlying architecture, producing fiscal strain and political fragility without restoring a durable, self-supporting loop.

There are plausible ways forward. They likely involve some combination of broadened ownership, redesigned tax bases suited to an economy where labor income is no longer central, and income mechanisms that are less tightly coupled to employment. But proposals should be judged by concrete criteria: do they keep aggregate demand sufficient to clear output, preserve incentives for investment and innovation, avoid unstable debt expansion, and remain politically sustainable over decades rather than election cycles?

This is where computational modeling becomes essential. Verbal reasoning alone quickly reaches its limits when a system is governed by feedback loops, delays, and nonlinear effects. Then ideology tends to fill the empty spaces. Models are always wrong in some respect, but a good model can be inspected, criticized, and improved. It can show where income comes from and where it goes. It can show how investment is financed, what happens as wages decline, and which policies move the system toward stability rather than crisis. It can reveal stresses and failure modes. Just as importantly, it can identify the policy regions where the system may become self-supporting again.

Current discussion of the economic impact of artificial intelligence remains largely qualitative and often proceeds one policy at a time. Universal Basic Income is a leading example. But the transition will not unfold one policy at a time; ownership patterns, fiscal flows, household balance sheets, demand, political arguments, and institutional adaptation will interact throughout.

An intuitive account of the transition and quantitative modeling are both needed.

This paper provides that intuitive account. It describes the transition as a connected process. It aims to clarify the underlying economic issues, show how a coordinated policy package may

address multiple stresses at once, and bring into view some of the political and institutional frictions that may obstruct adaptation even when workable policy paths exist.

The companion *New Loop Policy Architecture and Simulator* paper presents the quantitative simulation model. It describes a stock-flow-consistent representation of 20,000 households whose wealth, income, revolving debt, mortgages, and propensities to consume roughly correspond to distributions observed among families in the United States. It also includes firms, banks, and government as aggregate actors. In that model economy, automation gradually increases the substitution of capital and software for labor. Over time, value added shifts from wages toward profits.

The underlying question is simple to state, but difficult to answer: what forms of ownership and policy architecture can keep aggregate demand, investment incentives, and social stability within a viable range as automation advances?

## THE MODEL TELLS A STORY

This transition will not arrive as a single shock. It will unfold unevenly across sectors over ten to twenty years, punctuated by surprises and accelerated by feedback. So the model tells a story, not a moment.

**Early on, call it the "first wave,"** not a year on a calendar, machine intelligence stops being scarce. It becomes a commodity input: embedded in software, wrapped into workflows, and eventually integrated into physical systems. Then the decisive issue is no longer whether automation is possible, but how quickly it diffuses, and what it does to the economy's income architecture.

The first visible break is in information work, not because the work disappears, but because the same output can be produced with far fewer people. Accounting and compliance become "continuous" rather than seasonal. Routine analysis compresses into automated pipelines. You still see humans in the loop, but the loop is narrower than expected and narrowing further. Productivity rises, headcount stops rising, and then quietly falls.

A partner at a large professional firm says it plainly at an industry dinner: *"We used to sell hours. Now we sell outcomes. And outcomes don't need headcount."*

Automation of the physical economy follows, slowly at first, then faster, though never as fast as in the information sector. Early robotics are specialized: systems designed for narrow tasks. But the trajectory is toward generality, toward machines that can be retrained and redeployed, because the bottleneck is no longer the actuator but the cognition that coordinates perception, planning, and control. Automation seeps into warehousing, manufacturing, mining, agriculture, construction, maintenance, logistics, and then into the parts of those domains that used to be protected by "situational complexity."

Governments and firms alike recognize that automation offers a way to overcome the labor-cost advantages that had earlier shifted so much production into developing economies and hollowed

out much of the industrial base of the developed world. But this does not mean a return to the old employment bargain. China is automating aggressively and building the infrastructure to support it. This drives down the cost of industrial capability, so others feel compelled to follow. *"We had better do it too!"* becomes both a boardroom conclusion and a national policy instinct.

The result is that reindustrialization, where it occurs, needs far less labor than the industrial recoveries of the past. Output may return; broad hiring does not. Pressure on wages and employment therefore intensifies rather than eases.

**By the "second wave,"** firms begin to treat hiring as the expensive, irreversible decision. Managers defer hiring even when business is strong, because the option value of automation is rising. Displaced workers often find work, but less often find work that pays like the jobs that disappeared. Wage growth softens, then the wage share begins to slide.

At the same time, capital demand rises. Automation is not free. Even when cognition becomes cheap, building the intelligent equipment and software infrastructure needs up-front investments in hardware, sensors, integration, robotics, data pipelines, security, and the organizational work of changing how production is done. And because the advantage is competitive, speed matters. Firms borrow to transform. Banks and markets accommodate, until they start to notice that the same technologies raising productivity are also reshaping purchasing power.

Now the story becomes macroeconomic.

A strange squeeze develops. Households attempt to maintain consumption while labor income weakens. Some draw down savings. Some borrow. Revolving credit expands in the places you would expect: the middle and lower middle, where a loss of hours or wages is not a "temporary inconvenience" but a balance-sheet event. Firms borrow to finance automation investment. The economy's ability to produce improves, but the economy's ability to pay for what it produces starts to weaken and will collapse unless something replaces wage-linked purchasing power at scale.

Prices respond to opposing forces. Productivity and competition push prices down, especially in tradable sectors. But markup capture can push the other way, especially in concentrated markets or where switching costs are high. The model does not assume a single outcome. It lets pass-through and capture contend, because the policy regime must survive either a deflationary transition, an inflationary one, or a mixed reality.

Either way, debt service becomes the fragility channel. If the regime tilts deflationary, nominal incomes soften while nominal debts remain fixed. If it tilts inflationary, households face higher consumer prices while wages lag. In both cases, the upper tail cracks first. The warning sign is not the average household; it is the ninety-fifth percentile of debt-service pain. Lenders begin to see a pattern: this is not a normal cycle, and not a one-time shock. It is structural. It is the automation transition.

A bank risk officer puts it in a sentence that makes the room go quiet: *"The collateral is fine. The borrowers aren't. And it's not because they're irresponsible—it's because the income architecture is changing."*

**Then comes the recognition moment.** Credit tightens where it most hurts demand, and the economy becomes demand-constrained even while productive capacity is growing. The system can still produce, but fewer people can buy.

This is where the model begins: not at the start of automation, but at the point where the feedback loop between output and purchasing power starts to break.

The design problem is now explicit. If wages are no longer the dominant channel by which households receive income, what can replace the cash flow that supports mass consumption and social stability, without destroying incentives to invest and innovate?

The model's answer is not a single lever. It is an architecture: a set of rule-based stabilizers that shift purchasing power and ownership in ways that can scale with automation.

The first move is to broaden ownership of productive assets gradually, credibly, and with a hard cap. The model introduces a Public Trust Fund held on behalf of citizens, not an operating arm of government. The fund is structured as a trust, not a government agency, because property law is highly developed and can help protect the fund from political interference. Its purpose is not to run companies; it's to accumulate equity ownership and distribute part of the resulting returns. The trust activates only when measurable stress crosses a threshold, so it behaves like an automatic stabilizer rather than a permanently expanding institution.

When the trigger condition is met, the trust can be kick-started with bank-created credit. The bank lends to the trust; the trust purchases a slice of outstanding equity; households receive the cash proceeds; and the trust repays the loan over time. That launch mechanism matters because it accelerates broad ownership without requiring an immediate tax grab or a politically fragile appropriation.

But the core of the ownership transition is not the launch. It is the ongoing mechanism: what the model treats as a form of equity-dilution tax that is implemented as new share issuance allocated to the trust until an explicit cap is reached.

The obvious question is: why would corporations accept that?

They accept it in this story for the same reason they accept many things that look costly in isolation: because the alternative is worse, and because the rule can be made predictable, limited, and compatible with capitalist incentives.

A CEO of an automation-heavy manufacturer says it bluntly in an earnings call, after an analyst presses on "why tolerate dilution at all?": *"We can build capacity faster than the economy can absorb it. Customers don't come from spreadsheets. If broad purchasing power collapses, we're not 'more*

*efficient'. We're stuck with stranded capability. A stable rule we can price is better than a political firestorm we can't."*

That is the hard constraint: in a mature automation transition, the binding constraint is not productive capacity. It is demand. Firms cannot monetize productivity if the consumer base is insolvent or purchasing power is too concentrated. The dilution rule is framed as a demand-preservation instrument: it converts a fraction of the automation dividend into distributed purchasing power without requiring firms to raise wages for jobs that no longer exist.

A CFO phrases the acceptance logic in the language markets understand: *"If it's universal, capped, and rule-based, it's not 'expropriation', it's a cost of doing business in a high-automation economy, like a tax, like regulatory compliance, like stock-based compensation. The market can price it. What markets can't price is unpredictable backlash."*

That predictability is central. A small, stable, widely applied dilution rate is something investors can incorporate into valuation. It reduces the risk premium associated with tail events: windfall confiscation, arbitrary sector-specific punishments, or emergency interventions that arrive precisely when credit is already tight. A rule-based mechanism is not "free," but managers can read and understand it, and legibility lowers the probability of chaotic, discretionary policy.

A portfolio manager adds the quieter point capital markets rarely say out loud: *"The real threat to returns isn't a few basis points of annual dilution. It's the social legitimacy gap. If the system stops working for most people, the political system will eventually reach for something much cruder."*

The mechanism is also framed as a substitute for more distortionary tools. As labor income becomes less central, the tax base must shift. Governments will come after profits, consumption, wealth, or all three. In that political reality, a capped dilution mechanism can be presented as growth-compatible: it is gradual; it doesn't depend on brittle enforcement against profit-shifting; and it builds broad claims on capital without requiring repeated political battles over explicit confiscation.

The cap matters here. The trust is not allowed to become an unlimited accumulator of control. The policy goal is to create a stabilizing ownership share, not to nationalize the corporate sector. The cap makes the mechanism credible to founders, boards, and long-horizon investors: it ensures the rule does not converge toward takeover. And the trust's role is structured as passive ownership in economic intent: broad claims on returns, not politicized management. Firms accept dilution more readily if they can believe the trust is not a steering committee.

Under those conditions, the equity-dilution rule reads less like punishment and more like a premium paid for macroeconomic stability and continued license to operate in a society where labor's role in income formation is shrinking. It's a structured way for firms to buy what they need: solvent consumers, political legitimacy, and a predictable policy environment in which long-term investment remains rational.

The trust fund accomplishes another important goal: it addresses the duration-matching requirements of citizens' income funding needs over normal human lifecycles. In today's world, people are born without assets, although they may inherit some wealth eventually. When families are young, they need money raise children, fund college educations, buy a house. These large financial demands get in the way of saving. Students in college need income support. And in old age we have Social Security, but its implementation in some ways a pyramid scheme built on the idea that the young will finance the old. None of this works in an aging, deflating economy. The trust fund is a national endowment, a property trust owned by the citizens themselves, with a perpetual horizon.

The second pillar in the story is a broad consumption tax, modeled as a VAT wedge between producer and consumer prices, paired with a targeted low-income-household rebate so the tax does not function as a regressive demand-killer. A VAT is attractive in this world because it scales with consumption in an economy where wages are no longer the primary taxable flow, and because it can be enforced more reliably than an ever-more-complex wage-tax system. The model does however include an income tax (flat rate on wages above a minimum income), and an indexed corporate tax that rises somewhat as wages fall.

The third pillar is universal income support, defined in a way that matches the systems-engineering problem. In the model two different forms of income support may be chosen to fill the gap between market incomes and a baseline real-income pool. In different ways, both stabilize purchasing power rather than merely moving nominal transfers around. Both methods have a ratchet: once it rises, it does not fall. That is a realistic political assumption as much as an economic rule: benefits that can be withdrawn easily are not a stable foundation for an economy undergoing structural labor displacement.

Funding is handled as a hierarchy rather than a slogan: some share can be financed by monetary issuance; then trust balance contributes; then government balance contributes; with further issuance as the final backstop. The model is not asserting that issuance is consequence-free. It is making the backstop explicit so the model can test how much is required, when it is required, and what happens if it is used too aggressively.

These mechanisms work together to support one existential objective: keep the economy operating in a viable region as automation advances. They are parameterized rules whose job is to preserve the feedback loop that makes a modern economy function: production, incomes, consumption, investment, and social legitimacy.

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If this story is right, the payoff is enormous. The automation transition fades from emergency into baseline. Productivity gains appear as rising real living standards rather than as a collapse in employment-linked income. The economy continues to invest, invent, and build because consumers remain solvent and demand remains sufficient to clear output.

Even if the story is wrong, a quantitative model of these policies is still useful because it makes failure modes visible. You can see where the loop breaks, which instrument fails first, and what kinds of adjustment might restore stability.

But whether the story so far is right or wrong is not the only question. It's still a narrow tale, even a parochial one. How might different countries reach their own resolutions? What would it take? Artificial intelligence and advanced automation are global phenomena. Countries and regions will find their own paths forward, so there is also a global story…

## SAME STORM, DIFFERENT SHIPS

The hardest part of adapting to intelligent automation was that, while the old political and economic rules and reflexes no longer worked, each major power or region was compelled to make its own mistakes. Old habits were hard to break.

The storm hit everyone, but the ships' hulls were built differently. China had the engines: manufacturing scale, engineers, and command capacity. It also had a glacier of debt that deflation could crack like ice. The United States had the laboratories and the venture capital, but not the steering wheel. Its politics could argue for years about who deserved the gains even as the gains arrived on schedule. Europe had the deepest social contract and an instinct to keep people whole, but historically paid a price in speed and coordination.

As for the rest of the world, people who did not invent these machines could buy them. In theory, that meant they could compress history: import the tools, skip intermediate stages, and become consumer societies faster than the old powers ever could. In practice, it depended on who owned the machines, who collected the rents, and whether new productivity was converted into household purchasing power rather than siphoned upward and outward. The same technology that could build a middle class in ten years could also hollow out the middle class in five.

### CHINA

China entered the AI era with the world's greatest manufacturing base, a deep bench of engineers, and a state apparatus that could concentrate astonishing resources. But as the first hump arrived and automation began to hit information-sector wages and early-career employment, Chinese youth unemployment was already in the high teens.

China also carried an enormous debt stack, especially tied to property, much of it fixed nominal. In a deflationary environment, that is balance-sheet poison.

Land-transfer income and other property-linked revenues had long been a core funding channel for local government, especially at the prefecture, county, and city level, and that channel had been shrinking for years. In a deflationary regime, the local fiscal model and the debt model fail together.

Meanwhile, China's domestic consumer economy remained comparatively underdeveloped, so exports still played an outsized stabilizing role. Yet across the developed world, re-industrialization increasingly meant automation and capital deepening, not a wage boom, and therefore not an expanding consumer base for imported goods.

Beijing tried to redirect exports toward faster-growing regions, but ran into strategic pushback and geopolitical friction, especially in the Americas. And with weaker domestic consumption, "tax the consumer to keep the macro pump primed" had less leverage than it did in economies where consumption was the dominant engine.

Then there was Taiwan: smaller, wealthier, and institutionally more flexible, and owned the most strategically indispensable export category of all: advanced semiconductors, the engines of intelligent machines. Beijing couldn't ignore the risk that a widening performance gap between Taiwan and the Mainland might become a legitimacy problem at home. But forceful unification risked a severe U.S. reaction. Waiting for American semiconductor self-sufficiency might reduce that risk, yet the speed of AI diffusing into the knowledge economy compressed the timeline.

Also, taking Taiwan Semiconductor Manufacturing Corporation and Taiwan's entire semiconductor industry offline might offer a grim strategic advantage: it could delay global AI industry progress while giving China a chance to catch up with ASML and TSMC in semiconductor manufacturing technology. Yet that devilish gamble would leave a clear field for Korean and American companies, and it might be a long, long time before the world would forgive and forget. The costs to China itself could easily add up to trillions of dollars.

There was no clean solution, only choices trading one kind of instability for another. Ultimately, China's limited internal flexibility got in the way of transforming itself quickly. It could push forward most effectively by continuing to do what it already knew how to do.

China's timeline had been shaped by age, career, and succession, and by the perceived need to hand a unified China to the next leader. But events moved too fast. Internally, the system was simply not yet ready for its citizens to be wealthy without being occupied by hard work, and the risks of military force against Taiwan were too great.

Of course, China was endlessly clever about *applying* AI in their economy and their systems of political control. But with so much still depending on exports, China softened its image and adjusted its outward posture, not out of conversion but necessity. It refocused its export machine on products that helped regional neighbors in Asia and partners across Africa modernize faster. Those customers needed electric power equipment, networks, transport, and factory automation. These markets would give China a second chance in the coming global cycle of physical-sector automation, but it felt like a bitter second prize.

## UNITED STATES

America sailed into the storm from the opposite direction. It had the laboratories, venture capital, software talent, and cultural permission to move fast and break things. It could turn ideas into companies faster than anyone else. That was its superpower and its trap. The gains arrived first as private property, not as public design.

When automation began to compress wages beyond a few industries, the old postures and arguments awoke on cue. Conservatives insisted that displaced workers would simply find new jobs, as if history were a law of physics. Progressives wanted the winners to pay, as if punishment could substitute for a working economic engine. Neither side would admit what the numbers showed: in a mature, automated economy, labor could no longer be the primary conduit carrying purchasing power to households at scale.

Washington regarded AI as an essential domain of power and control, and tried all the familiar old policy compromises. Borrow through the transition. Stimulate on the way down, patch on the way up. Argue about taxing wealth. Squeeze health care. But the leak was structural, so patchwork could not hold. When wage income stops rising, household balance sheets stop rising with it. And when household balance sheets stall, the collateral behind the entire credit system begins to soften.

The Federal Reserve understood deflation, but it could not cut interest rates below the basic fact of missing cash flow. Monetary policy could change the price of short-term credit, but it could not guarantee the incomes that made credit safe. The stress appeared where it often does in America, first in revolving debt, then in housing, where the middle class is levered. Mortgages began failing in ways that did not look like the last crisis, and housing prices began to fall, trapping families in homes they could not afford. The payment shocks were not driven by teaser rates or exotic underwriting. They came from ordinary families whose work had been made cheaper than their debt, if not eliminated.

As the mortgage markets creaked and groaned, the problem revealed its second face. Corporate credit began to wobble too. If consumers can't buy, revenues can't hold. If revenues can't hold, debt can't roll forever. The storm that began in software reached the balance sheets of the physical economy.

Only then did the argument change. It stopped being a fight about fairness and became a fight about continuity. The question was no longer about the morality of capitalism, but how it could be mechanically stable when labor was no longer the main distribution system. The crisis forced a realization that should have been obvious in retrospect. Capitalism had to be preserved by redesigning its loop, so that household purchasing power could be sustained without relying on ever-expanding labor income.

Perhaps most remarkably, it was AI itself that made the necessary policy solutions credible and acceptable. Political arguments are usually qualitative, self-serving, and incomplete. But AIs that could cross-check each other, and could reason over deep new streams of economic data, made it possible to understand the problem quantitatively. Human specialists and intelligent machines described and debated the modeling using vast arrays of detailed, real numbers from the real economy.

More important, AIs made that understanding accessible to everyday people. All the work on "AI safety," reducing hallucinations, and building medical systems smarter than most physicians paid off in an unexpected way. Machines had become more trustworthy, more transparent, and in many domains more trusted than politicians.

Machines that could explain complex matters simply and honestly provided the revelations that transformed politics. Their avatars debated human politicians fluently, in real time. The smartest politicians allied themselves with machines, not to manipulate voters, but to inform voters at their own levels, often personally. This didn't end propaganda, but it changed the balance. Mark Twain is often credited with saying, "A lie can travel halfway around the world while the truth is putting on its shoes." Now the truth, running barefoot, began to outpace the lies.

## EUROPE

Europe entered the storm with three primary weaknesses.

First, Europe was not a single ship. It was an armada of different histories, interests, and institutional designs. Coordination was slow, and each capital faced its own version of the transition dilemma. The core European Union remained very different from more nationalist and illiberal states on its periphery, whose instinct was to seek local advantage over continental coherence. Meanwhile, the Ukraine war shifted the balance of industrial and military relevance eastward, raising the strategic weight of Poland and, later, of Ukraine after accession. Germany, for all its industrial prowess, no longer stood quite so naturally at the center of the European story.

Second, the euro and its governing bureaucracy had been built above all to prevent inflation, Germany's oldest economic fear, not to confront deflation. The European Central Bank was not well designed for the kind of emergency adaptation that the new transition required. The question was whether Europe could act together quickly enough, or whether national authorities would, in practice, be forced to improvise around the edges of the common system.

Third, Europeans tended to regard disruption as a social hazard rather than an opportunity. European political systems were built to preserve legitimacy. Demography also weighed on growth. When work and wages thinned, the first question was not "who failed?" but "what must be stabilized here?" And stability, almost by definition, is not naturally forward-looking.

On the other hand, Europeans did have tools. They had mature consumer economies and value-added tax systems already in place. They had long experience with the welfare state, wage support, and reducing working hours. They had universal health care systems, and publics accustomed to bargaining over the boundary between life and work. In a sense, Europe had been rehearsing for decades the idea that working time is negotiable, even if that had something to do with the continent's long habit of lagging the technological frontier.

The very fact that Europeans could speak openly about reducing working time gave them a head start in the argument that mattered. They understood that the real issues were how to preserve demand, how to preserve dignity, and how to keep people in their homes if mortgages began to destabilize.

## THE SAINT-GUILLAUME MEETING

Even when Gaul was a Roman province, its tribes were proudly and stubbornly independent. During the Second World War, Charles de Gaulle sustained the myth, and then the reality, of French independence from a London apartment by little more than his own abrasive force of will. After the war, he gave France a new constitution and a nuclear force de frappe, both as instruments of power and as symbols of national independence and dignity. He amputated French Algeria in the same spirit, believing that the new France could not preserve its identity while dragging a colonial appendage behind it.

That same tradition left modern France unusually well equipped for the AI transition. It possessed both the administrative machinery and the intellectual elite needed to operate it. Of all the European states, France was best prepared not merely to endure the transition but to design and guide it.

It was the French who invented the value-added tax in the first place. Though the British coined the phrase "political economy," its modern expression as *dirigisme*, a government-guided economy, was quintessentially French. France's technical elite, trained to think in systems, supplied a steady stream of administrators able to translate policy into operational reality. France's constitutional architecture, bequeathed by de Gaulle, contained levers for decisive action when needed and legitimacy for necessary action. And the French people, however fiercely they argued over details, were already familiar with the idea that leisure and working time are political variables. For them, the compass heading was not completely strange. It was recognizable.

The conceptual roots of the French plan had many parents and had been discussed qualitatively and piecemeal around the world. But they were integrated, made specific, and clearly articulated in the form we know today at a four-day meeting convened by the French President at *Sciences Po* in Paris. That is the common, and not always affectionate, name for the Paris Institute of Political Studies, one of France's most prestigious and distinctive institutions. More independent than most French universities, and supported by its own foundation, *Sciences Po* was never merely a school of political science. Its curriculum mixed history, law, economics, sociology, and international relations, and its highly international student body was required to spend a year abroad.

The conference invitees came mostly from *Sciences Po's* sister institutions in French public policy, economics, law, and administration, with guest speakers and observers from around the world. Representatives of the European Central Bank and national bankers attended. Notably, no American business executives were invited.

The President of France was a young and energetic man, recently elected in part because he had openly and empathetically acknowledged the worsening economy. He opened the conference by noting indicators of economic distress that were rising in France and globally, due to rising automation and the increasing flow of profits to capital instead of labor. Then, to emphasize the gravity of the situation, he read aloud the first four articles of the Declaration of the Rights of Man and of the Citizen. That is the revolutionary text that still stands behind French constitutional tradition.

1. Men are born and remain free and equal in rights. Social distinctions may be founded only upon the general good.
2. The aim of all political association is the preservation of the natural and imprescriptible rights of man. These rights are liberty, property, security, and resistance to oppression.
3. The principle of all sovereignty resides essentially in the nation. No body nor individual may exercise any authority which does not proceed directly from the nation.
4. Liberty consists in the freedom to do everything which injures no one else; hence the exercise of the natural rights of each man has no limits except those which assure to the

> other members of the society the enjoyment of the same rights. These limits can only be determined by law.

"I say to you," he concluded, "that our task here is not to take from the rich and give to the poor. Nor is it to invent a wholly new society. Though we meet at *Sciences Po*, our task is not primarily political. It is economic, rooted in changing patterns of capital flows. Machines are replacing human labor, and capital flows reflect that. Therefore, whatever might be your personal politics, this is a crisis of capitalism. We are not here to replace capitalism but to preserve it, because capitalism emanates from rights of property and free association, and from the law.

"So this is the question before us: How may all citizens not merely envy the products of machine intelligence and automation, but afford and enjoy their benefits, and live more securely and more freely under the rights proclaimed in our Declaration? This is what we are here to answer."

## *LA NOUVELLE BOUCLE*

The Saint-Guillaume Meeting, as it is now known, produced a document in three chapters: *Diagnosis*, *Treatment Plan*, and *Health and Wealth*. The President asked that it be written for a wide readership rather than as a technical memorandum. It appeared under the title *La Nouvelle Boucle* (The New Loop), and France became the first major country to implement the program it set forth.

### DIAGNOSIS

*Diagnosis* explained that the shrinking base of jobs and wages was not an ailment that could be treated directly. It would not be sufficient in the long run, and likely not even possible, to create enough new jobs to replace all those lost to automation. Nor could the state simply tax companies to replace the lost wage income, because consumption would weaken along with labor income. With inadequate consumption, firms could not sell enough to provide an adequate tax base.

The second part of the diagnosis was even more unsettling. Companies do not exist primarily to create jobs. They exist foremost to create wealth for those who own them. Jobs are a means and an expense, not a corporate purpose. Companies that kept unneeded workers indefinitely would become competitive failures.

The deeper problem, contradicting centuries of political habit, was not that companies and their owners were accumulating too much wealth. It was that too few people owned sufficient equity, and therefore the public did not share adequately in the wealth that companies created. As machine intelligence and automation became the dominant sources of productivity and wealth, a society in which most citizens lacked any meaningful claim on capital would become mechanically unstable. A core aspect of the disease, to say it again, was that the public did not own enough equity.

The third factor was that there were, in the end, not many durable secrets about artificial intelligence. The underlying technical knowledge was available to anyone who could read and build. Restrictions on the most advanced chips could slow diffusion at the frontier but would not prevent it. In knowledge-based industries especially, most competitive advantages would be fleeting.

Without durable moats, prices in sectors touched by machine intelligence would be driven steadily downward. The result would be persistent deflation.

In the context of French social and political history, this recital of facts was revolutionary; but data presented at the conference demonstrated the truth of it. The shock was somewhat tempered by similar data showing that the entire developed world was beginning to experience the same slow chill.

## TREATMENT PLAN

*Treatment Plan* began from a simple proposition: the old economic loop, which circulated output, wages, demand, and investment, now suffered from a progressive condition that could not be patched. Since wages were becoming an ever-narrower channel for both consumption and taxation, a new foundation had to be built beneath both. That foundation must sustain household purchasing power, provide an adequate tax base for the State, and still leave firms with enough profit and reinvestment capacity to remain dynamic.

In other words, though we redesign the macroeconomic channels, we cannot abolish circulation. Circulation reveals whether we got the redesign right. The New Loop might distribute purchasing power through new and different mechanisms, but value must keep moving through the whole system. Households must still be able to buy. Firms must still be able to sell. The state must still be able to tax. Capital must still be able to reinvest. If any major channel collapses, the whole organism weakens. The economic heart must still pump and perform the same circulatory work, even if some vessels must widen because others are shrinking.

It was a Finnish woman who stated the point with memorable precision: when the remodeling was complete, the future economy should still be circulating roughly the same amount of money. Any viable solution would require that. Yet this obvious constraint also carried an implicit reassurance. A river may be redirected into different channels, she said, but the amount of work the flowing water can do remains the same.

Do this properly and the system could recover its balance and sustain itself. The transition need not become a mechanism for arbitrary loss on one side and windfall gain on the other. Given the productivity gains made possible by intelligent automation, it should be possible to preserve social balance while leaving society as a whole better off.

The prescription therefore fell into four main parts, each addressing a distinct failure in the Old Loop.

First, it had to replace the shrinking wage-income channel. Second, part of the capital stream had to be redirected into broad ownership and public revenue. Third, debt deflation had to be prevented from breaking households and banks. And fourth, critically, tax policies had to remain simple enough to be understood, stable enough to be trusted, and fair enough to be sustained politically.

**Replace the shrinking wage-income channel**. Wages have traditionally done two jobs at once. They have provided households with purchasing power, and they have provided governments with a tax base. Jobs would not vanish altogether, but wage income would no longer be broad enough to carry the full weight of either function. Household income therefore had to be supplemented directly.

Working groups broke out to debate two proposals.

The more familiar was a Universal Basic Income, set at a meaningful level, for example near the bottom third of family incomes in the pre-transition economy. It would provide a stable floor beneath consumption, and with it a measure of dignity and social participation.

The UBI could be funded from three sources. The first was a value-added tax on all consumption, without exception, paired with a rebate for households below a minimum income threshold. The second was dividend income from a new Public Trust Fund proposed by a separate working group. The third was a modest contribution from general government revenues. In the event of a shortfall, the UBI would be supplemented by limited government money issuance. Under the more deflationary conditions created by intelligent automation, such issuance was expected to be modest and not inflationary.

The value-added tax would have to be substantial, on the order of 15 to 20 percent, but its burden would be offset in two ways. First, lower-income households would receive the rebate. Second, although the UBI itself would not be taxed, all other income would be taxed at a low flat rate above a protected untaxed minimum. In this way, the system would support household purchasing power while keeping the overall tax structure broad, intelligible, and economically efficient.

But the German delegates in particular, and European Central Bank observers as well, had reservations about a UBI. Despite Germany's strong history of social provision, many felt that a basic equivalence principle was at stake: support on that scale should not be detached too abruptly from labor contribution. Since that argument was politically awkward in a world of steadily declining wage income, the objection was instead framed in fiscal and monetary terms. A UBI would be very expensive at the outset. It would absorb a large share of tax revenue before the new sources of revenue had fully matured. It would also be strongly stimulative, driving consumption higher and stirring old German anxieties about inflation.

The real dispute was not over whether household income had to be supplemented, but over how ambitious the new instrument should be at the start. Another working group therefore set out to design a less expensive, more incremental policy, which they called the Universal Income Stabilizer, or UIS. Its purpose was simply to fill the national wage gap as it emerged, while preserving the universal character that a full UBI might eventually require. A UIS could later be reshaped into a UBI if necessary. But if the broader transition program succeeded, the long-run practical difference between the two might prove smaller than either side first assumed.

German and European monetary observers tended to prefer the stabilizer form, seeing it as narrower, more conditional, and therefore more credible as a transition instrument. A politician

may say: *if the problem is inadequate demand and falling prices, why be timid?* A central banker or finance ministry figure replies: *because credibility still matters, and the state should not begin by promising more than the machinery can support.*

UIS looked like a bridge. UBI looked like a destination. With strong arguments on both sides the conference could not immediately resolve the question, so the Treatment Plan offered both options. Different countries might make different choices. Either way, the old wage channel would be supplemented by a broader and more deliberate circulation of purchasing power.

**Broaden public equity ownership and public revenue.** The second set of policies was designed to address the existing concentration of capital without destroying the incentives for innovation and entrepreneurship. The aim was simple: if fewer people were going to earn their living mainly from wages, then more people had to receive part of the nation's income as owners.

A Public Trust Fund, owned by the people, would be established for that purpose. In effect, the nation would plant a public orchard inside the private economy. It would acquire a bounded grove, large enough to bear real fruit, but still clearly short of control. It would not seize the whole forest. This was not wholly novel in France, which already had its *Fonds de réserve pour les retraites,* or FRR, invested in market assets.

The fund would begin with a substantial but limited acquisition of equity. The banking system, backed by the government, would lend it a sum equal to 10 percent of the outstanding equity of existing French corporations. That loan would be used to purchase shares in each corporation in proportion to its existing equity. Dividends from those shares would be used to repay the loan. This repayment would somewhat delay the benefits, but still give the public an immediate foothold in the productive wealth of the country.

At the same time, companies would be required to issue new dilutive shares to the trust fund, equal to 2 percent of their equity at the beginning of each fiscal year. But this process would be capped. By law, the fund could never own more than 49 percent of the total equity of French companies, with the expectation that it would not need to rise above roughly 25 to 30 percent. After the original acquisition loan was repaid, dividends flowing to the fund would become a continuing source of funding for universal income.

This would be a kind of wealth tax, but not a direct personal one. It would restructure some capital ownership without abolishing private property or private control. In this way, the public stake would widen gradually rather than through confiscation, growing along with the corporate sector itself.

European Union law required that the fund be neither a tool of the government nor accessible as an asset by the government. The fund could be insulated by making it a property trust operated for the benefit of the citizens, as opposed to a Sovereign Wealth fund governed under the Santiago Principles. Even so, those Principles provided guidance and management form. France could not enforce dilutive share issuance upon companies domiciled in other countries, so any international holdings would be purchased outright.

The corporate tax system would also be redesigned. Undistributed profits, after depreciation and genuine capital reinvestment, would be taxed. Distributed dividends, by contrast, would be taxed only in the hands of recipients, not at the corporate level. Dividends passing through the Public Trust Fund would remain untaxed there as well. These policies would encourage dividend payouts.

A second tax change would tie the corporate tax rate to the overall national decline in wage income. As labor's share of national income fell, the tax rate on profits could rise correspondingly. A firm that became more productive through intelligent automation, while contributing to a shrinking wage base, could therefore afford to pay more tax on its profits. But corporations could also reduce that burden through further productive investment and through dividend distributions. The tax system, in other words, had to follow the new river of value.

Over time, that reinvestment would generate more wealth, higher productivity, and larger dividend streams, some of which would flow through the Public Trust Fund and the UBI or UIS to support consumption.

Within the EU it is easy for a company to move its headquarters and tax reporting to a different country. The OECD's Two-Pillar Solution for cross-national taxation could begin to address the problem of capital flight from equity dilution and remodeled corporate taxation, but the *Treatment Plan* acknowledged that more would be required to fully address capital flight.

**Protect both households and lenders from debt deflation.** Long-term fixed debt, such as mortgages, requires special attention in a deflationary economy. Prices and wage income can fall even as real economic output rises. If prices fall and wages follow, a household's fixed mortgage payments become an increasing share of the monthly budget. In time, that becomes insupportable. Meanwhile the bank receives a windfall, until the mortgage fails, because the fixed nominal payment is worth more in deflated euros.

The solution was innovative and subtle. It introduced a kind of pressure regulator. By law, the original contract payment would become indexed to a hybrid measure reflecting both the price level and household income. If the economy deflated, the required mortgage payments would shrink. But the bank would still be made whole, because the government would neutralize the lender's nominal loss by paying the difference directly to the bank.

The mortgage would remain a real obligation, still a contract. But it would no longer be a rigid conduit through which falling prices and falling wages could freeze prudent households in place, force them into default, and then transmit the shock onto bank balance sheets.

This new policy was easy to defend to citizens, to the banks, and to the French State itself. It would preserve household solvency, as well as the credit system on which savings, lending, and investment depend. The duty of the state was not to side with either the creditor or the debtor, but to regulate the pressure between them so that both could pass through the transition intact.

Many French households were renters who would see no benefit from mortgage protection, but *Treatment Plan* suggested that universal income support at the proposed level would help such

households directly. Rents might be “sticky” and not contract much with deflation, but they would be unlikely to rise and should eventually adjust.

**A fair and simple tax system.** The conference recommended a new tax system conceived in concert with the new economic loop. The authors argued that simplicity would make the long transition into this new economy easier to measure and harder to exploit or manipulate politically.

The elements of the tax proposal were these. Universal income support would not be taxed at all. Value-added tax would be applied in the familiar way, but rebated to low-income households, so that the rebate phased out as income rose. Income tax would be levied at a flat rate on all income above a set minimum, with that minimum indexed to deflation. The corporate tax rate would be tied to a hybrid index of prices and declining wages. The principle was simple: as companies eliminated labor, they could afford to pay a higher rate, and the revenue would help support consumption.

## HEALTH AND WEALTH

The conference understood from the outset that this transition would be long, perhaps twenty years or more. Emergency relief or a temporary correction would not be enough. France would need a social contract that could endure and stabilize a prolonged passage into a more automated economy, while remaining politically legitimate, economically intelligible, and transparently measurable.

This was neither socialism nor an attempt to replace the market. Companies would remain private, competitive, and strongly oriented toward profit and reinvestment. But because all citizens would collectively own the Public Trust Fund, the gains from corporate success would no longer flow only to a narrow class of private owners. They would also return, in part, to the citizenry as a whole.

That changed the political meaning of profit. In the old language, labor and capital stood opposed, and rising profitability could easily be seen as coming at labor’s expense. The new arrangement would soften that tension, because citizens would participate not only as workers and consumers, but also as owners. Ownership changes one’s perspective. Productivity, innovation, and profitability would no longer appear simply as corporate victories. They would strengthen a common national asset in which every citizen had a stake.

This new economic loop depended in part on dividend distribution, and the tax system was designed accordingly. Distributions would not be taxed at the corporate level, not as a gift to companies, but as an incentive for shareholders, among whom the public fund would become increasingly important. Universal income support, meanwhile, was more than a pension substitute. It would support households across the life cycle: in youth, in study, in child-rearing, in interrupted work, and in old age. And because the system strongly favored reinvestment for corporate growth, citizens would gain not only income, but wealth. Over time, the Trust fund itself would grow.

The authors, while delicately avoiding certain words, presented this as a cure for the old French habit of treating profit and social protection as opposing principles. If the tax system remained

simple, if dividend flow was encouraged, and if citizens came to see themselves as owners as well as workers, then growth and security would no longer stand against one another. They would become reinforcing elements of the same loop.

During the decades-long transition, the New Loop would be measured openly as it evolved. Progress could not simply be proclaimed; it would have to be tracked, explained, and adjusted as household income, prices, investment, dividend flows, tax receipts, and the national fund changed. Machine intelligence could help by turning large, messy data into usable forecasts and policy stress tests, and by presenting results in forms citizens could understand. In a transition of this length, trust would rest not only on outcomes, but on the visible, auditable integrity of the measurements.

European Union law imposed important design constraints on any French version of the New Loop. The Public Trust Fund could not be simply an arm of the state, exercising discretionary political control over private firms. It would have to be constituted as an independent and largely passive investment trust, governed by law and fiduciary rules rather than by ministers. Likewise, mortgage indexation and bank make-whole provisions would have to be framed as general financial-stability and social-cohesion measures, applying by rule rather than by favoritism. These constraints did not make the plan impossible, but they did require that it be designed and implemented with institutional discipline.

## AND THE CYCLE TURNED

Historically, the American Revolution came before the French one. The United States was first. But in this newer revolutionary age, France saw more clearly than we did what had to be done. France acted first, with a coherence, elegance, and élan that America lacked.

There were rough passages, of course. As the simulations had predicted, the great deflation could not be entirely prevented, and economic policy moved, as it always does, with a lag. Ten years passed before deflation began to stabilize and turn around, and ten more before that process fully played out. Yet the larger fact remained: it worked. The transition was not painless, but it held, and the world that emerged was better and fairer than what might so easily have followed.

The Public Trust grew into a triumph. The decision not to tax distributed dividends at the corporate level proved especially important. It encouraged payout, enlarged the Trust's income, and helped return a larger share of rising productivity to households as purchasing power. The value of the Trust grew, then kept rising, and that visible growth gave everyone a tangible stake in the new system. Consumption strengthened, firms found real markets for what they produced, and the New Loop began to reinforce itself.

The tone of politics now reflected the fact that everyone felt wealthier and more optimistic, particularly younger households. Income inequality eased greatly, though a wealth gap remained. The bottom moved up. Households with great wealth did not lose much of that wealth. As the designers predicted, neither share dilution nor the wage-linked corporate tax became confiscatory. The fixed dilution rate was modest, and over time it was overtaken by compounding gains in productivity and profitability.

Ordinary households became much less fragile. Families started saving earlier and could invest more confidently in education. Universal income support did not end work, but for many it was a gift of freedom. People could pursue ideas, create new businesses, and start families with more confidence. Some could choose work more for its meaning and less purely for its necessity. For those so inclined, work became less a matter of compulsion and more a matter of vocation.

These changes made capitalism stronger. Consumers remained solvent. Demand remained real. Firms kept investing because markets still cleared and the system remained investable. Innovation could still be highly rewarded. But the economy no longer depended so heavily on fragile households, debt expansion, and chronic insecurity. Incentives were better aligned. Ownership was broader. Social stability was stronger.

With less wisdom, it might not have turned out that way. But it did.

*March 7th, 2026.*

This future history is one possible scenario. We can't know in advance the speed or full extent of the transition. Artificial intelligence and advanced automation will move faster in some sectors than in others. Bottlenecks, delays, and political resistance are also likely. Many indications suggest that the geopolitical backdrop, especially around Taiwan, may prove less benign than the one assumed here.

But uncertainty is exactly why we need computational modeling. A parameterized, adjustable framework makes it possible to examine different rates of change, different depths of impact, and different policy responses directly.

I began this project about six weeks ago. Three of those weeks were spent writing and polishing this scenario and its companion paper. Today the New Loop model consists of 7,200 lines of Python code. I wrote none of those lines myself, though I closely guided and tested their development incrementally, and discussed details with several AIs along the way. The project does include one utility module, 863 lines long, that I wrote in C# 12 years ago. OpenAI's Codex translated it into Python flawlessly in a few minutes.

ChatGPT prepared deep-research reviews of this paper in about six minutes. Claude also generated a deep review which I consider to be the better of the two. It took me much longer than that to study those reviews and think through their implications.

Three months ago, in December, this project was not yet feasible. That is information-sector automation happening in real time.

At the outset, it was not evident where this work might lead. It was expeditionary. Revolutions and great wars have emerged from much less disruption than what artificial intelligence may bring, but the modeling leaves me optimistic. There are plausible paths forward that need not leave workers behind. Capitalism can become stronger and more resilient, not abolished. It is not inevitable that wealth and power will concentrate still further while impoverishing most of society.

A crisis, if it comes, could create the opening for a long-needed political and economic reconstruction.

Politically, the transition could be as long and hard as Shackleton's journey out of Antarctica. Expect endless crippling argument. Conservatives will cling to inherited structures. Progressives will want to expropriate wealth. Corporations and wealthy stockholders would fiercely resist dilution, even if it is slow, measured, and capped. Some countries, shaped by different histories, institutions, and civic habits, will navigate this transition better than others.

Still, my sense is that the New Loop is structurally sound and directionally correct. I spent much of my adult life investing against inflation, and there may yet be more of that. But intelligent automation is likely to be deflationary. Increased productivity will reduce the use of human labor and the share of revenue directed to wages. Lower incomes will depress consumption and make sales more sensitive to price. Every country and company will have access to essentially the same AI capabilities, so competition will also press prices downward. Developed countries will use intelligent automation against the traditional labor-cost advantages of developing economies.

Deflation can be slow and exhausting, or devastating. People displaced by automation will often find it very hard to find new work. For many older workers, it may be impossible. But the hard fact is that no modern economy can remain healthy by shifting everyone into marginal personal services.

The cash flows of economies everywhere will be rewired, but will that rewiring be opportunistic? Or will it be deliberate? Today we can't know. America's Social Security system was built on the assumption that current workers support retirees through current wage taxation. It is already under strain. Large-scale job losses from intelligent automation would break that structure.

Nor will it be possible for the federal government simply to spend its way out of the dilemma. Wealth taxes will not solve it either. There is not enough accessible wealth for that to work at the required scale, and untethered wealth taxation would damage saving itself. It would push people who had acted prudently over a lifetime into unnecessary dependence.

I hope this work has been worth your time, and that some readers will continue with the companion paper, *New Loop Policy Architecture and Simulator*, and explore the New Loop simulator. The intelligent machine transition stands to become the most consequential economic and political challenge of our time. It deserves deeper, more integrated thought than it has yet received.

Roger Ison

**To preserve the storytelling nature of this paper, additional reading references have been placed at the end of the companion paper.**